SNOWFLAKE DATA ENGINEERING PROJECT

Healthcare Intelligence Hub

Patient, Cost & Care Analytics

Phase 4 — Project Planning Phase

Institutional template-based phase document

Student	Aashi Bindal
Enrollment No.	0827CI241004
Team ID	[Team ID]
Institution	Acropolis Institute of Technology and Research · Department of CSIT
Faculty Guide	[Faculty Guide]
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Prepared from the verified project source (query.sql, Health/streamlit_app.py, Health/snowflake.yml, six CSV datasets). Bracketed fields are placeholders for information not supplied.

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Phase 4 — Project Planning Phase

Project: Healthcare Intelligence Hub: Patient, Cost & Care Analytics **Template covered:** Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points), 8 marks

> **Dates and effort.** The project files contain no commit history, timestamps or sprint records. Calendar dates are therefore shown as **relative days (estimates)**. Story points are planning estimates. "Completion evidence" refers only to artefacts that exist in the repository. Fill the bracketed fields with your actual dates before submission.

4.1 Project Planning Template

Field	Value
Date	18 October 2026 *(as printed on template)*
Team ID	[Team ID]
Project Name	Healthcare Intelligence Hub: Patient, Cost & Care Analytics
Maximum Marks	8 Marks

4.1.1 Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Table 4.1 — Product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Planning & Requirements	T-1	Study problem, profile six CSVs (row counts, keys, value domains)	3	High	[Member 1]
Sprint-1	Planning & Requirements	T-2	Ideation and requirement documents (Phases 1–2)	5	High	[Member 2]
Sprint-1	Data Access & Integration	USN-1	Create `HEALTH_DB` and the RAW, STAGING, CURATED and MART schemas	2	High	[Member 3]
Sprint-1	Data Access & Integration	USN-2	Define `CSV_FORMAT` and `HEALTHCARE_STAGE` ; create six typed RAW tables; load CSVs	5	High	[Member 3]
Sprint-2	Data Processing	USN-3	Build STAGING tables (TRIM, INITCAP, NOT NULL keys)	5	High	[Member 4]
Sprint-2	Data Processing	USN-4	Build five dimensions and `FACT_PATIENT_VISITS`	5	High	[Member 4]
Sprint-2	Business Analysis	T-3	Create ten MART KPI views	3	High	[Member 1]
Sprint-2	Data Processing	T-4	Validation queries: row counts, NULLs, duplicates, orphans	2	High	[Member 2]
Sprint-3	Business Analysis	USN-5	KPI cards: visits, patients, cost, satisfaction, readmission %, LOS	3	High	[Member 1]

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Business Insights	USN-6	Monthly trend and visit-type donut	2	High	[Member 2]
Sprint-3	Business Insights	USN-7	Facility performance chart and table	3	High	[Member 3]
Sprint-3	Business Insights	USN-8	Doctor leaderboard	2	Medium	[Member 4]
Sprint-3	Business Analysis	USN-9	Cost breakdown by visit type	2	High	[Member 1]
Sprint-3	Business Analysis	USN-10	Insurance and patient payment totals (KPI view)	1	Medium	[Member 2]
Sprint-3	Business Insights	USN-11	Readmission rate by diagnosis	2	High	[Member 3]
Sprint-4	User Interaction	USN-12	Sidebar filters: date range, facility, specialty, visit type, diagnosis category	5	High	[Member 4]
Sprint-4	Data Update & Reporting	USN-13	Package and deploy app (`snowflake.yml` , `pyproject.toml`)	3	Medium	[Member 1]
Sprint-4	Testing	T-5	Data-quality, transformation, KPI and dashboard tests (Phase 6)	4	High	[Member 2]
Sprint-4	Documentation	T-6	Final report, README and demonstration preparation (Phases 7–8)	3	High	[All]
Backlog	Security	USN-14	Roles, grants and masking policies	5	Low	—

4.1.2 Project Tracker, Velocity & Burndown Chart (4 Marks)

Table 4.2 — Project tracker (relative schedule; actual dates to be filled in)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)	Completion evidence in repository
Sprint-1	15	6 Days	Day 1 [date]	Day 6 [date]	[actual]	[date]	`CREATE DATABASE/SCHEMA` , `CSV_FORMAT` , `HEALTHCARE_STAGE` , six `*_RAW` tables in `query.sql` ; CSVs present. *Load command not scripted.*
Sprint-2	15	6 Days	Day 7 [date]	Day 12 [date]	[actual]	[date]	STAGING, CURATED and MART sections and validation queries in `query.sql`
Sprint-3	15	6 Days	Day 13 [date]	Day 18 [date]	[actual]	[date]	KPI, chart and table code in `streamlit_app.py`
Sprint-4	15	6 Days	Day 19 [date]	Day 24 [date]	[actual]	[date]	Filters in `streamlit_app.py` ; `snowflake.yml` ; `tests/local_validation.py` ; this documentation

Velocity:

With a 6-day sprint and a planned velocity of 15 story points per sprint, the team's average velocity (AV) per iteration unit is:

> AV = Sprint velocity ÷ Sprint duration = 15 ÷ 6 = **2.5 story points per day**

Across 4 sprints the plan covers 60 points. If the actual completed points differ, recompute AV as (sum of completed points ÷ number of sprints) ÷ 6.

Burndown Chart: A burndown chart is a graphical representation of work left to do versus time.

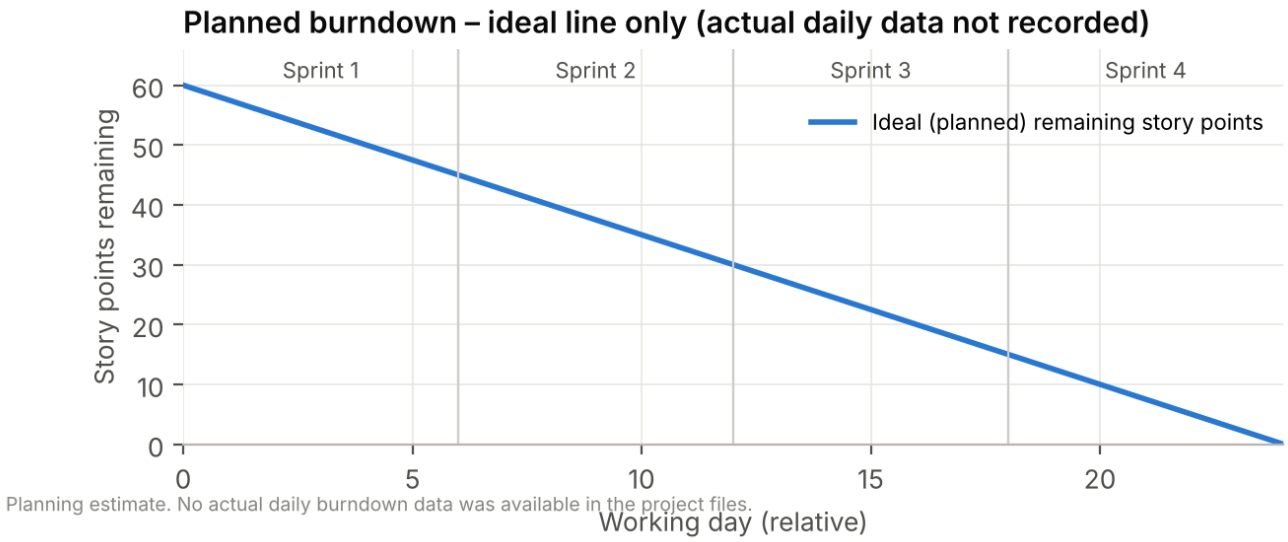


Figure 4.1 — Planned (ideal) burndown for 60 story points over 24 working days. No actual daily progress data was recorded, so only the ideal line is shown. Plot your actual remaining points against it.

References (from template): Atlassian Agile guides on project management, epics, sprints, estimation and burndown charts; Visual Paradigm: Scrum burndown chart.

4.2 Work Breakdown Structure

```
1  Healthcare Intelligence Hub
├── 1.1 Initiation & Ideation ........ problem statements, empathy map, brainstorming
├── 1.2 Requirement Analysis ......... FR/NFR, DFD, user stories, tech stack, journey map
├── 1.3 Design ....................... architecture, layers, star schema, app wireframe, security
├── 1.4 Data Engineering
│   ├── 1.4.1 Database & schemas
│   ├── 1.4.2 File format, stage, RAW tables, data load
│   ├── 1.4.3 STAGING cleaning
│   ├── 1.4.4 CURATED dimensions & fact
│   └── 1.4.5 MART KPI views + validation queries
├── 1.5 Application .................. KPI cards, 8 charts, 2 tables, 5 filters, deployment
├── 1.6 Testing ...................... DQ, transformation, KPI, dashboard, performance
└── 1.7 Documentation & Demo ......... report, README, demo script, presentation
```

4.3 Phases, Milestones and Deliverables

Table 4.3 — Milestones (relative timeline, estimated)

#	Milestone	Planned (relative)	Deliverable	Dependency	Status (evidence)
M1	Ideation complete	Day 2	Phase 1 document	—	Documented
M2	Requirements baselined	Day 4	Phase 2 document	M1	Documented
M3	Design approved	Day 6	Phase 3 document, diagrams	M2	Documented
M4	Data loaded to RAW	Day 6	RAW tables populated	M3	DDL present; load not scripted
M5	Curated model built	Day 12	STAGING, CURATED, MART	M4	Present in `query.sql`
M6	Dashboard feature-complete	Day 22	`streamlit_app.py`	M5	Present
M7	Testing complete	Day 23	Phase 6 report, test scripts	M5, M6	Local tests executed; Snowflake tests Not Executed
M8	Documentation and demo	Day 24	Phases 7–8, README	M7	This set

Critical path: M2 → M3 → M4 → M5 → M6 → M7 → M8. Every analytic output depends on the curated fact table.

4.4 Resource and Tool Requirements

Resource	Purpose
Snowflake account, warehouse `COMPUTE_WH`	Storage, compute, Streamlit hosting
Snowsight	Worksheets, file upload, app editor
Snowflake CLI (optional)	`snow streamlit deploy` using `snowflake.yml`
Python 3.11 + pandas (local, optional)	`tests/local_validation.py`
Git / GitHub	Version control and publication
Graphviz, Markdown	Diagrams and documentation

4.5 Roles and Responsibilities

Team composition was not supplied. Assign names to the roles below.

Role	Responsibilities	Assigned to
Team lead / coordinator	Planning, tracking, integration, demo lead	[Name]
Data engineer	Snowflake objects, loading, STAGING and CURATED SQL	[Name]
Analytics engineer	MART views, KPI definitions, validation queries	[Name]
App developer	Streamlit dashboard, filters, deployment	[Name]

Role	Responsibilities	Assigned to
QA and documentation	Test cases, report, README	[Name]

4.6 Risk Assessment and Mitigation

Table 4.4 — Risk register

ID	Risk	Likelihood	Impact	Mitigation	Status
R1	Snowflake trial credits expire	Medium	High	Use X-Small warehouse with auto-suspend; keep `query.sql` re-runnable	Open
R2	Manual data load is repeated, causing duplicate RAW rows	Medium	Medium	`CREATE OR REPLACE` RAW tables; duplicate-key checks in STAGING	Mitigated by design
R3	Inconsistent category labels	Low	Medium	TRIM/INITCAP in STAGING	Mitigated, but INITCAP side effects found (Phase 6)
R4	App slow on large data (full table into pandas)	Low now / High at scale	Medium	Cache (`ttl=600`); future SQL push-down	Accepted for current volume
R5	PII exposure	Medium	High	Proposed roles and masking	Open
R6	Team availability	Medium	Medium	Clear role ownership; documented scripts	[to update]

4.7 Testing and Documentation Plan

Activity	When	Owner	Output
Local replica tests (`tests/local_validation.py`)	After each SQL change	QA	Console report / JSON
Snowflake validation (`tests/snowflake_validation.sql`)	After full rebuild in Snowflake	Data engineer	Results recorded in `TEST_CASES.md`
Dashboard functional tests (manual checklist)	Before demo	App developer	Screenshots, pass/fail
Documentation	Throughout; final in Sprint-4	All	`docs/` , README, PDFs

4.8 Progress Tracking and Completion Criteria

Track progress in the tracker table (4.1.2) and the burndown chart. The project counts as complete when:

1. Every High-priority story in Table 4.1 has its acceptance criterion met (Phase 2, Table 2.4).
2. `tests/snowflake_validation.sql` has run in Snowflake with results recorded.
3. The app opens in Snowsight and all five filters work.
4. All eight phase documents and the README are committed to the repository.